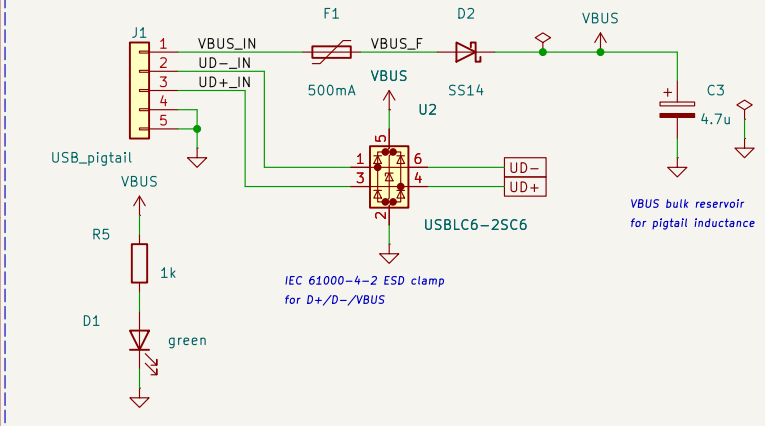
USB INPUT + PROTECTION

solder pads for USB-A pigtail to the Pi
(VBUS, D-, D+, GND, shield->GND)

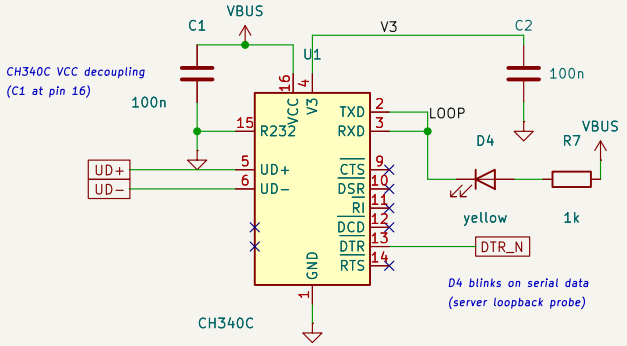
protects the Pi USB port from
board or hood-wiring faults

reverse-polarity guard for a
miswired pigtail (~0.3V drop)



power LED current limit (~3mA)
USB power indicator

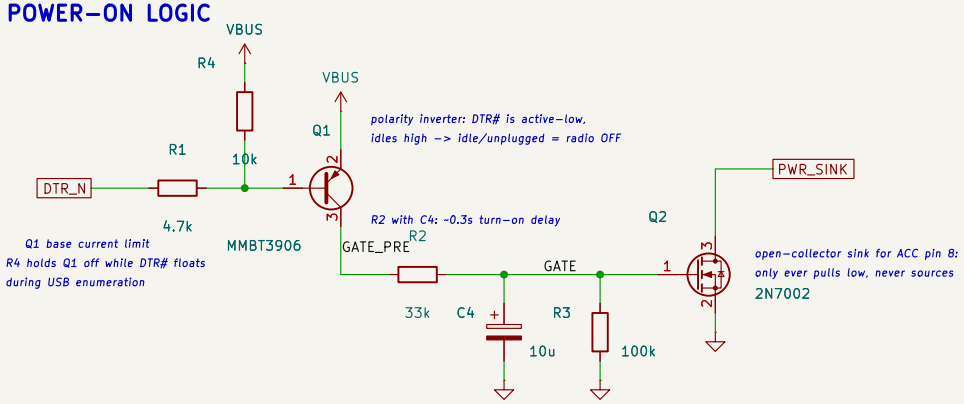
USB-SERIAL BRIDGE



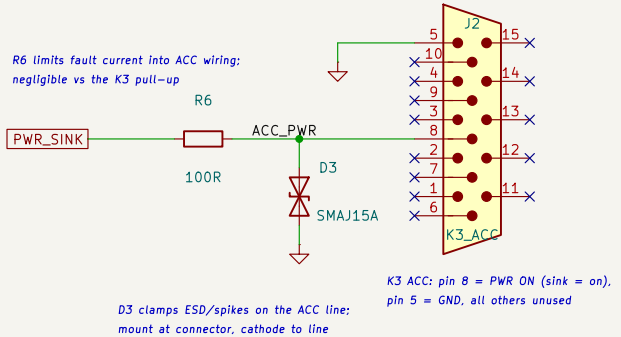
USB-serial bridge: only DTR# is used
TXD->RXD loopback lets the server
probe that the dongle is present

C2: V3 internal-regulator bypass (per CH340 datasheet)

POWER-ON LOGIC



K3 ACC OUTPUT



Fail-safe: unplugged/unpowered/port-closed = radio OFF
CH340C DTR -> PNP inverter -> RC -> 2N7002 open collector -> K3 ACC pin 8
W6JY

Sheet: /
File: k3pwr-dongle.kicad_sch

Title: k3pwr USB power dongle

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/1